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## Research Agenda

My research studies how AI systems can transform supply chain decision-making under uncertainty, spanning three interconnected thrusts: (1) agentic retrieval and reasoning over heterogeneous enterprise data, (2) reinforcement learning for operational pricing and routing, and (3) governance and auditability in AI-augmented procurement and logistics. The first thrust develops multi-agent architectures that retrieve, reconcile, and synthesize evidence from procurement documents, supplier master data, and contract repositories distributed across vector, relational, and graph databases. I investigate hybrid routing strategies that combine LLM reasoning, embedding similarity, and domain heuristics, with confidence-gated handoffs and bounded verification loops to address routing confusion, retrieval gaps, and incomplete evidence. This work is grounded in a supplier-intelligence orchestration platform deployed in a multinational healthcare and consumer goods context and extends through a collaboration with a Tier-1 hyperscaler to construct agentic BOM knowledge graphs that produce reproducible, joinable parameter tables for downstream optimization. The second thrust applies deep reinforcement learning and dynamic programming to operational decision problems in freight and maritime logistics. Current projects include dynamic pricing for containerized barge services on the Mississippi River, where I compare policy-gradient methods against static baselines under stochastic demand, and market-sizing frameworks that combine ensemble machine learning with geographic, transit-time, and commodity filters to quantify serviceable demand for inland waterway freight. This line of work connects to broader questions about modal innovation, decarbonization through waterway utilization, and how learned pricing policies can accelerate adoption of underutilized freight corridors. The third thrust, which crosscuts the first two, examines how agentic AI systems can maintain auditability, reproducibility, and governance compliance in high-stakes enterprise settings. I study deterministic ingestion pipelines augmented by agent-assisted extraction, read-only query architectures with strict access controls, and evaluation frameworks that assess correctness, latency, and provenance. This thrust reflects a conviction that trustworthy deployment, not only model performance, is the binding constraint on AI adoption in procurement and supply chain operations.

## Education

Massachusetts Institute of Technology (MIT), Cambridge, MA — 2026 – Present
PhD in Transportation

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Massachusetts Institute of Technology (MIT), Cambridge, MA — 2025 – 2026
MST in Transportation

Massachusetts Institute of Technology (MIT), Cambridge, MA — 2023 – 2024
Master of Applied Science in Supply Chain Management

University of Illinois at Urbana-Champaign, Champaign, IL — 2021 – 2023
Master of Business Administration

The University of Texas at Austin, Austin, TX 2011 – 2016
Bachelor of Arts

## Research Experience

Massachusetts Institute of Technology, Center for Transportation and Logistics (Deep Knowledge Lab)
2025 – Present
Research Assistant

- Building agentic AI systems that enable procurement-focused agents to retrieve, reconcile, and synthesize evidence from RFx documents, supplier master data, and contract repositories distributed across heterogeneous data silos and database types.
- Designing end-to-end workflows for supplier intelligence, including document understanding and structured extraction, retrieval over multi-format artifacts, and tool-using agents that generate auditable outputs for sourcing, compliance checks, and contract term comparisons.
- Developing reinforcement learning and dynamic programming methodologies for operational decision problems, including optimal routing in maritime and over-the-road settings under realistic constraints (capacity, service times, and cost structures).

## Publications

- "A Multi-Filter Market-Sizing Framework for Mississippi River Containerized Freight: Assessing Serviceable Obtainable Demand through Ensemble Machine Learning," *Transportation Research Record*, March 2026 (under review).
- "Fast Barge, Smart Prices: Deep Reinforcement Learning for Mississippi Container Service," *Annals of Operations Research*, Special Issue on Innovations in Maritime and Port Logistics: Optimization and Data-Driven Decision Making, January 2026 (under review).
- "Deep Reinforcement Learning for Mississippi Container Service," *INFORMS Transportation Science and Logistics Society Conference*, July 2026.
- "Optimizing Drop Trailer Networks: Advanced Analytics for Efficient Supply Chains," *World Conference on Transport Research*, July 2026.
- "Strategic Investments in Freight Decarbonization: Quantifying the Benefits of Zero-Emission Corridor Deployment and Strategic Rail Modal Shifts," *World Conference on Transport Research*, July 2026.
- "Assessing the Total Addressable Market of Barge Service on the Mississippi River," *Transportation Research Board*, Inland Waterways Operations and Infrastructure, January 2026.
- "Enhancing Sentiment Classification of E-commerce Reviews for Actionable Insights Using Large Language Models," *IEEE*, September 2025.
- "Drop Trailer Management in Volatile Networks," *Supply Chain Management Review*, June 25, 2025.
- "Drop Trailer Forecasting in Volatile Networks," *Procedia Computer Science*, Elsevier. Proceedings of the 15th International Conference on Ambient Systems, Networks and Technologies (ANT), April 2024.

## Capstone Work

- "Drop Trailer Management in Volatile Networks," MIT Supply Chain Management Capstone Research, MIT DSpace, May 2024.

## Talks and Presentations

- "AI-powered Automation in Ports," Workshop on AI-powered Automation in the Ports, Rutgers University (DIMACS Center), Piscataway, NJ, March 2026.
- "Assessing the Total Addressable Market of Barge Service on the Mississippi River," Transportation Research Board Annual Meeting (TRB 2026), Washington, DC, January 2026.
- "From Silos to Agentic Systems: Leveraging Agentic AI for Supplier Intelligence," MIT Center for Transportation and Logistics Agentic AI Roundtable, December 2025.
- "Enhancing Sentiment Classification of E-commerce Reviews for Actionable Insights Using Large Language Models," IEEE High Performance Extreme Computing Conference, September 2025.
- "Advanced Analytics for Efficient Supply Chains," 7th Interdisciplinary Conference on Production, Logistics and Traffic, Darmstadt, Germany, March 2025.
- "Drop Trailer Management in Volatile Networks," SCALE Expo, MIT, January 2024.

## Teaching Experience

- SCM.256 Data Science and Machine Learning for Supply Chain Management, Spring 2026, Teaching Assistant.
- Research Mentor, GCLOG Capstone, MIT CTL, July 2025 – December 2025. Provided technical and methodological guidance on problem formulation, pricing models, and empirical analysis for the student research paper "Monetizing Modal Innovation: Pricing Next-Generation Barge Services."
- AI in Supply Chain and Logistics Management, MIT CTL Summer School, July 2025, Teaching Assistant.
- SCM.256 Data Science and Machine Learning for Supply Chain Management, Spring 2025, Industry Partner Project Sponsor.

## Professional Experience

Amazon, Tokyo, Japan Jun 2024 – Aug 2025
Senior Operations Manager, Amazon Logistics (AMZL)

- Drove an 11% QTR increase in TPH, processed 110,000+ PPD as site leader of 300+ associates, and achieved #1 ranked site metrics (Q3 & Q4) for JP Super Region 3.
- Directed implementation of an AI-driven sortation process, reducing mis-sorts by 9% and saving a projected 11,500 labor hours annually across multiple cycle shifts.

- Streamlined inbound, sortation, and pick workflows, improving average WIP compliance by 10% and contributing to a 5% improvement in PDD rates for outbound deliveries.
- Launched comprehensive safety and productivity initiatives, culminating in a 15% reduction in incident rates and boosting employee engagement scores by 22% from Q2 to Q4 2024.
- Collaborated with academic partners and internal research teams on data-driven process improvement and applied machine learning initiatives.

Flock Freight, San Diego, CA — Jan 2022 – Aug 2023
Senior Analyst, Operations & Analytics

- Managed dynamic pricing model, providing real-time quoting for 60%+ of revenue. Optimized cost and margin estimates for probability-based pooling calculations, responsible for cost accuracy, conversion optimization, and margin maximization.
- Improved costing model accuracy by 10% (MAPE) through tenure, representing a $6MM YoY improvement in cost savings and a $4.2MM increase in gross profit.
- Automated pricing processes, resulting in a 40% increase in productivity and allowing strategic focus on higher-level pricing strategies.
- Collaborated cross-functionally with teams in engineering, data science, and product management to implement the APE ML pricing engine, resulting in streamlined automation of pricing processes and a 40% decrease in labor hours.
- Constructed and executed batch rating quoting automation tools for enterprise customers, removing workload equivalent of 4 QA associates.
- Directed daily pricing audits with Customer Success and Sales teams, assessing historic conversion, gross profit, and cost savings data to determine unique customer-specific strategies.

Zillow, Seattle, WA — Jun 2021 – Dec 2021
Pricing Analyst, Acquisitions

- Led valuation of $200MM+ in residential real estate utilizing data from internal and external datasets. Analyzed Zillow pricing valuations and risks, leveraging detailed financial models for strategic decision-making.
- Controlled Austin and San Antonio market research to enhance departmental accuracy. Utilized tools such as Tableau, Excel, SQL, Python, and proprietary ML tools.
- Monitored market data, analyzing real-time pricing information for weekly market intelligence reports; enhanced team accuracy (MAPE) by 2.7% through tenure via outlier analysis within the sales funnel.
- Executed 5 adopted proposals within Sales, Data Science, and Market (Austin) specialization groups to revamp efficiency metrics, resulting in an 18% reduction in case time per analyst.

Arrive Logistics, Austin, TX — Feb 2017 – Jun 2021
Pricing Analyst, Strategy

- Developed and performed company pricing strategy through championing strategic and tactical price optimization; promoted best-in-class pricing strategy, helping increase contract revenue by 320% from Jan 2018 to Jan 2021.
- Led multiple deals simultaneously ranging between $1MM to $30MM, closing $150MM+ in revenue through tenure.

- Maintained in-depth knowledge of industry trends, competitor analysis, and market dynamics, providing strategic insights to enterprise clients as well as internal stakeholders.

## Skills

- **Methods:** Reinforcement Learning, Dynamic Programming, Mathematical Optimization, Retrieval-Augmented Generation, Knowledge Graph Construction, Multi-Agent Orchestration, Statistical Modeling, Forecasting, Discrete Choice Modeling
- **Tools and Frameworks:** LangGraph, Neo4j, Amazon Neptune, FAISS, Pinecone, ChromaDB, PyTorch, Stable Baselines3, HuggingFace Transformers, OpenAI API, Anthropic API, Gurobi, NetworkX, Weights & Biases
- **Programming:** Python (Pandas, NumPy, SciPy, Scikit-learn, Statsmodels), SQL, Git, LaTeX, Bash
- **Cloud and Infrastructure:** AWS (S3, EC2, SageMaker), Docker
- **Visualization and BI:** Matplotlib, Seaborn, Power BI, Tableau, Excel
- **Languages:** English (Native), Japanese (Intermediate), Spanish (Basic)

## Service and Community

MIT Science Policy Initiative, ExVD Member
Austin Oita Sister City Committee Member
Keep Austin Beautiful Volunteer
Japan Society of Boston (Member)

## Professional Memberships

INFORMS (Member)
Transportation Research Board (Member)
World Conference On Transport Research Society (Member)